A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Conditions for life on Earth - Set 1

Mark scheme

AQA spec 3.1.1; The living environment; textbook pp. 5-9

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Mars has low mass so weak gravity and could not retain a dense atmosphere **1** Very low atmospheric pressure means water would boil / evaporate away even at low temperatures **1** Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO_2, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil **1** Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured **1** Too close (Venus) - too hot – too far (Mars) - too cold **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface **1** The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun **1** Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted – the Moon's gravity keeps the tilt stable and creates tides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1 General physiological solvent:** most biological reactions take place between substances dissolved in water **1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes **1** High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) **1 Anomalous expansion:** ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4
02.2	**1 mark each** 1 mark for each of the 4 points below. **1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) **1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3 **1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water **1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels **1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion – waste products of one process are raw materials of another so wastes do not build up **1** Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere **1** Allowing more plant growth, further stabilising climate and soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	3
03.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Lack of ancient historical / instrumental data **1** Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions **1** Lack of sophisticated equipment / inability to measure many factors **1** Limited coordination between researchers – lack of data collection in many areas (oceans, poles) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell **1** Air bubbles trapped in ice give the composition of the past atmosphere (CO2, methane) **1** Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) **1** More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
05.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2.7 - 2.45 = 0.25 billion = 250 million years **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (million years) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans **1** Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
05.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Oxygen in the atmosphere absorbed ultraviolet light **1** A dynamic equilibrium of reactions between O, O_2 and O_3 (the ozone layer) developed **1** Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) **1** Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) **1** The lower CO_2 concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion **1** Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas **1** Regulation of atmospheric composition (eg oxygen / CO_2 by photosynthesis and respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Transpiration** → water lost as vapour from the leaves **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
06.1	1 mark for the point below. **1** Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Age / date of the sample (radioisotope dating) **1** Past temperatures **1** Composition of the past atmosphere (eg CO_2 / methane concentrations) **1 Temperature:** more magnesium is incorporated into calcite at higher temperatures **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present **1** Ocean currents redistribute heat and nutrients – changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** There was no oxygen / ozone layer so high levels of UV radiation reached the land surface – water absorbed the UV and protected organisms in the oceans **1** Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely **1** The atmosphere contained toxic gases such as ammonia – the early Archaea were anaerobic so did not need oxygen **1** Water acted as the solvent for reactions / transport **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first 3 - the question asks for 3, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	

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